

Mechanical Problem Set #5 - Energy Storage Mechanism

Executive Summary:

We designed a bistable mechanism that stores enough energy for a pencil to be launched 1.5 meters with maximum horizontal distance. We converged on a design that incorporates $n=8$ flexures with a starting angle $\theta = 18^\circ$, a thickness $h = 2$ mm, and a length $L = 80$ mm. Modeling each flexure as a pseudorigid body with two torsional springs, we optimized the flexures' thickness to length ratio, as well as the starting angle. Furthermore, we modelled the frame's side walls as simply supported beams when calculating the energy that the device as a whole stored.

Math Model:

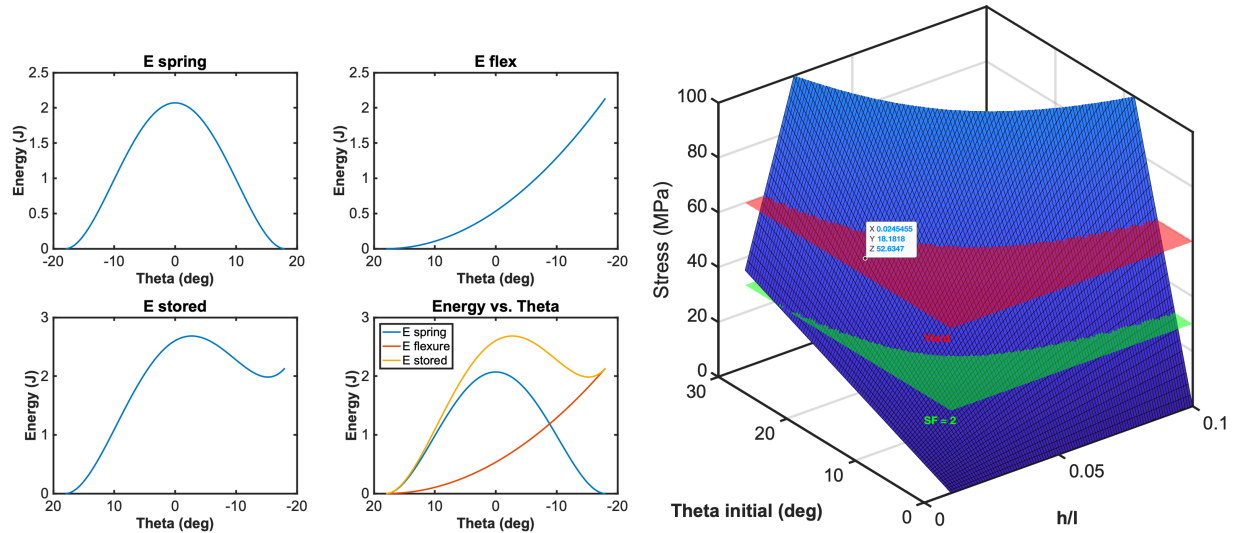
Initially, we modeled the flexures with PRBM as two torsional springs and one axial spring representing the compression of the flexures. However, the side walls of the mechanism were bending first. To improve our model, the axial stiffness was replaced with the stiffness associated with loading a simply supported beam (representative of the side walls); this model was most representative of the bistable mechanism's behavior in reality.

Using this model, energy stored in the mechanism, both in the simply supported beam (E spring) and the flexures (E flex), was plotted as a function of the displaced angle theta. As shown in the below graph to the left, the total stored energy (in yellow) has two stable equilibriums at the beginning and end of its motion. It is also observed in the model that the energy required to actuate the mechanism is lower than the energy required to load the mechanism, which matches our actual 3D printed mechanism.

We also considered a model where the frame walls were fully rigid, and the flexures were modeled as the combination of two torsional springs and a buckling beam. However, this model had multiple flaws, such as not being an accurate indicator of bistability and the energy storage for the model being too low.

We modeled energy losses from two sources: drag from the pencil moving through the air, and friction from the pencil sliding through the holder on the mechanism. From these losses, we calculated an upper bound for predicted efficiency of 32%.

To calculate stress, we used the formula $\sigma = Mc/I$. We created a surface plot of stress as a function of the initial starting angle, θ_0 , and the thickness to length ratio h/l to optimize both values. Ultimately, we decided on a θ_0 of 18 degrees and an h/l ratio of 0.025 because this puts our modeled stress at 52.6 MPa (slightly below the yield stress of 60 MPa). We are confident in this decision because we believe that the stress in our model is much higher than the stress of the actual mechanism due to two design changes. The added fillets between the side walls and the flexures would decrease the stress concentration at this interface, and the side wall cutouts would give the flexures more flexibility when the mechanism is actuated.



Verification Method and Results:

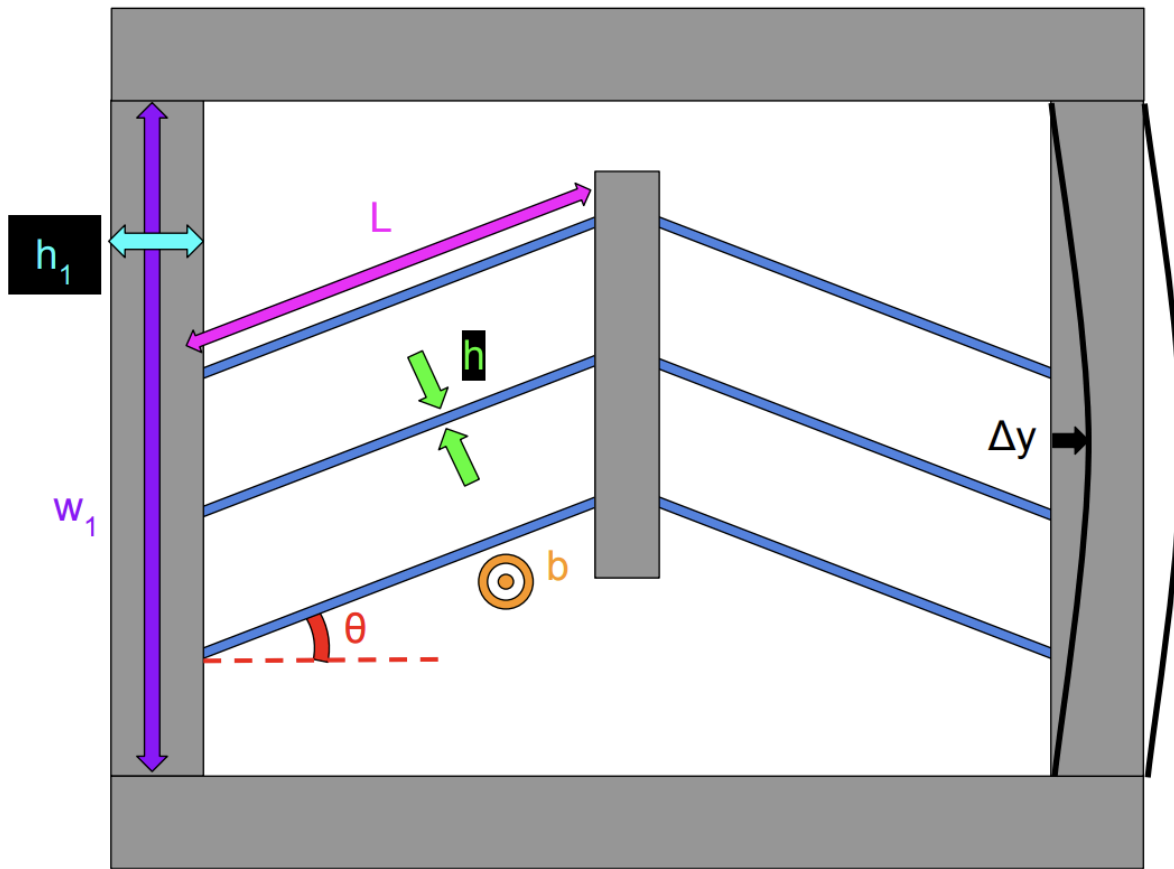
To test our designs, we launched a pencil from each of our mechanisms and videoed their launches to measure how they performed. Using slow-motion video, we compared the peak height of the pencil's trajectory to a blue piece of tape on the wall that indicated the 1.5m mark. For our final design iteration, we conducted 5 consecutive test launches and found that all of them cleared a height of 1.5 m. Their crashing points were within a 20 cm radius from each other, exhibiting a mean distance of 11.5 cm. To measure the energy transferred to the pencil, we shot a pencil while the mechanism was completely vertical, then measured the peak height of the pencil; calculating $E_{transferred} = m_{pencil} * g * h$ yielded 0.12 J. To calculate the mechanism's efficiency, we divided the transferred energy by the maximum energy predicted by our model. This resulted in an efficiency of 4.6%. The efficiency is lower than expected due to the nature of 3D printing, which leaves small gaps in the printed mechanism and would result in a lower Young's modulus. We also found that the energy storage and functionality of the mechanism is highly dependent on the print quality, with poorer prints corresponding to less energy storage.

Discussion/Learnings:

Our device went through many iterations of designs. We saw that bistability was hard to achieve with an open frame of three beams, so we made the frame closed to minimize how much the frame bends outwards, also helping us to store more energy in the device. Then, after consistently achieving bistability, we saw that small changes in angle, flexure thickness, and especially wall thickness made a large difference in the energy of our device. Another factor we learned to consider was the material properties of 3D printed PLA, as we either had to change the Young's Modulus for different infills of our prints, or print at 100% infill. After considering all of these factors in our design, we found that our physical device and our model were showing discrepancies in energy and stress. Stress discrepancies can be accounted for as our model did not take into account design choices we made such as fillets and slots in the frame next to the flexures.

Appendix

Geometric Parameters



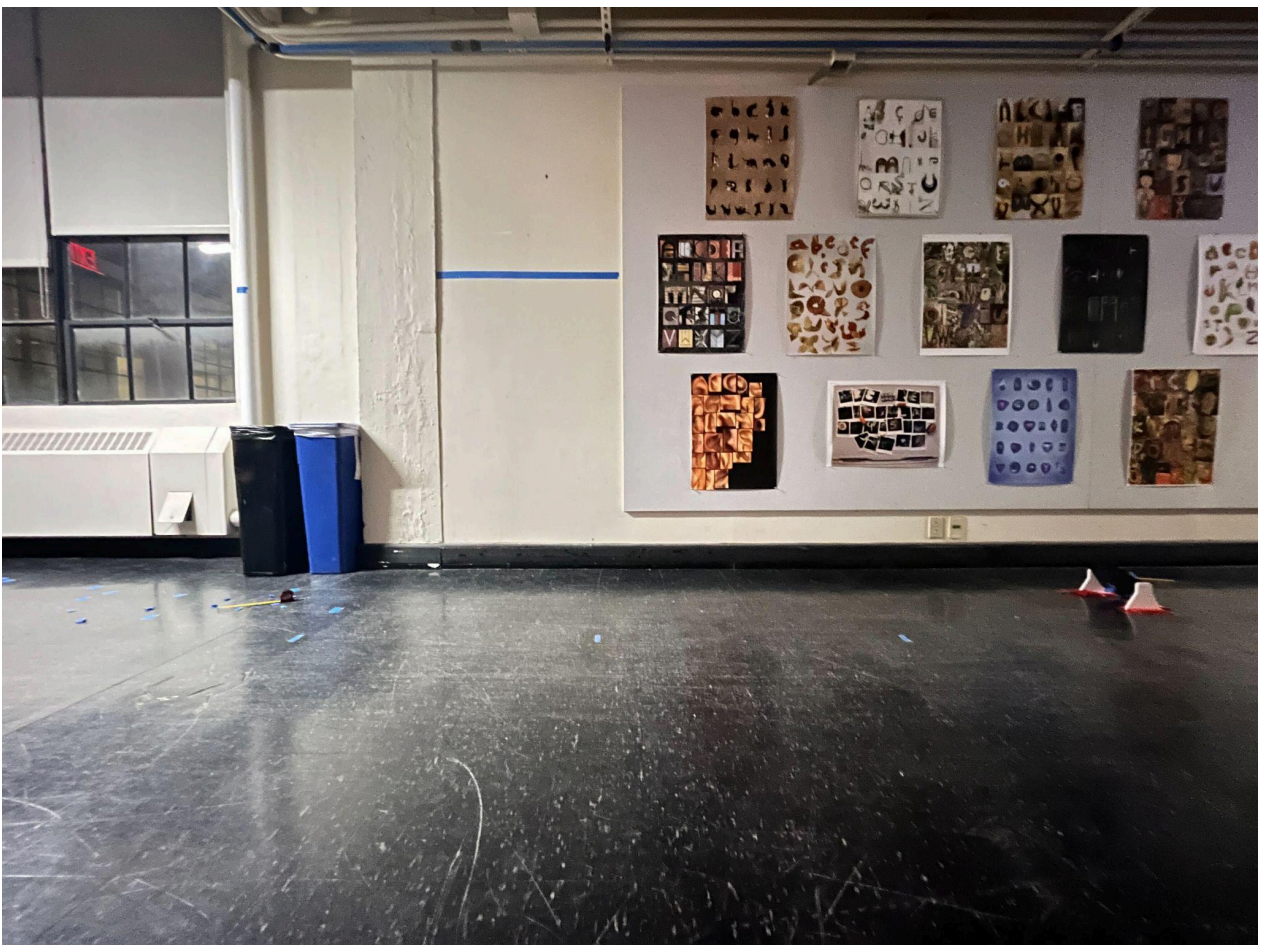
$n = \text{number of flexures}$

Functional Requirements

Variable Name	Symbol	Range	Justification	Testing
Minimum Energy Stored in Mechanism	E_s	0.365 J	$\frac{1}{2}m(v_x)^2 = 0.233 \text{ J}$ $mgh = 0.117$ <u>Energy Loss due to Friction</u> $F_f * l = 0.003 \text{ J}$ <u>Energy Loss due to Drag</u> $F_d * s = 0.013 \text{ J}$ <u>Energy Stored in Mechanism</u> $mgh + \frac{1}{2}m(v_{max})^2 + (F_f * l) + (F_d * s) = 0.365 \text{ J}$	2.68 J (from model)

Efficiency Upper Bound	η	0.32	Energy in pencil/energy in mechanism $\eta = \frac{E_{pencil}}{E_{pencil} + E_{flexures}} = \frac{mgh + 1/2m(v_{max})^2}{mgh + 1/2m(v_{max})^2 + E_f + E_d}$	0.046
Actuation Mechanism Force	F_A	$50 < x < 100 \text{ N}$	Enough force to avoid accidental deployments from moving the mechanism around, and less than the force required to turn a lever on an electrical box	80 N
Actuation Displacement	δ	$0.15 \pm 0.05 \text{ m}$	Shortest: half the length of the pencil To ensure the pencil is at an equilibrium position before launch (at least half the mass is on the mechanism making it stable) Longest: Length of the Print bed	0.04 m
Mass of Mechanism	m	$0.02 < x < 10 \text{ lbs}$	Needs to be able to be easily transported around campus, but needs to be at least the mass of the pencil to ensure that the pencil (10 g) can get launched, and not the mechanism	0.3 lbs

Testing Set-Up



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